

Education

Indian Institute of Information Technology, Lucknow

B.Tech in Computer Science and Business; CGPA: 8.13/10.0

Lucknow, India

Jul. 2024 – May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems (SQL, Transactions, ACID), Computer Networks, Operating Systems

Technical Skills

- **Languages:** Java, C++, C, Python, JavaScript, SQL
- **Core CS & Software Design:** Data Structures & Algorithms, Object-Oriented Design (OOD), System Design basics, Synchronous & Asynchronous Design Patterns, SDLC, Design Patterns, Distributed Systems Fundamentals
- **Backend & Databases:** Node.js, Express.js, RESTful APIs, PostgreSQL, MySQL, JWT Authentication, Database Transactions, Concurrency Control, Rate Limiting
- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **AI & LLM Integration:** Gemini API (LLM), Prompt Engineering, AI-Assisted Development, Real-Time AI Inference, Model Output Processing
- **DevOps & Tools:** Git, GitHub, Docker, CI/CD Pipelines, Jest & Supertest (Unit & Integration Testing), Load Testing, Linux, Postman, Cloud Deployment (Vercel, Render, Railway)

Projects

Transaction Ledger & Balance Management System

Node.js · PostgreSQL · React · Jest · Docker

[Live Demo](#) · [GitHub](#)

- Made an explicit concurrency-control trade-off in a multi-tier financial backend: implemented PostgreSQL row-level locking (SELECT FOR UPDATE with ORDER BY id) over naive table locking to eliminate deadlocks under concurrent transfers.
- Designed a double-entry ledger schema with balance_after snapshots for full audit trails; enforced request idempotency via PostgreSQL UNIQUE constraints so retried requests never double-process, a key reliability property for always-on financial APIs.
- Built a reconciliation engine running 3 automated SQL integrity checks (balance vs. ledger sum, debit-credit parity, stuck transactions); unit- and integration-tested atomic success, rollback-on-failure, and idempotent replay with Jest.
- Load-tested 20 concurrent transfers (54ms, zero monetary loss) to validate ACID compliance under load; automated schema migrations and test runs through a CI/CD pipeline on every deploy.

Interview Forge – AI-Powered Mock Interview Platform

React · Node.js · PostgreSQL · Gemini AI · Jest

[Live Demo](#) · [GitHub](#)

- Designed the backend architecture for an AI-native interview platform, orchestrating asynchronous calls to the Gemini 2.5 Flash LLM and a third-party code-execution API within a single request pipeline without blocking the client.
- Engineered structured system prompts constraining LLM behavior to single targeted questions; iteratively refined prompt design to control model output quality, prevent answer leakage, and maintain a realistic interviewer persona across sessions.
- Integrated glot.io API for sandboxed multi-language code execution (Python, Java, C++, JavaScript); added server-side timeouts and sequential test-case execution to keep the system reliable when an upstream dependency is slow or fails.
- Configured secure cross-domain HTTP-only JWT cookies (sameSite: none, secure: true) between Vercel and Render; added Helmet security headers, rate limiting, and Jest + Supertest integration tests with mocked dependencies for CI.

Competitive Programming

- **Codeforces:** Specialist, Max Rating: 1507. [Anish033](#) — strong in algorithms, dynamic programming, greedy, binary search
- **CodeChef:** 3-Star, Rating: 1628 [anish_033](#)
- **Problem Solving:** 800+ data structures and algorithmic problems solved across 50+ contests on Codeforces, CodeChef, and LeetCode covering arrays, trees, graphs, strings, and dynamic programming [LeetCode: Anish033](#)

Positions of Responsibility

- **Axios (Technical Society)** – Web Wing Member, IIIT Lucknow Aug 2025 – Jul 2026
- **Eifer (Sports Society)** – Senior-Member (organised inter-house tournaments with 200+ participants), IIIT Lucknow Present